

Kombinatorika

Izmed N slučajnih, neodvisnih dogodkov ima vsak n_i izidov. Število različnih izidov sestavljenega dogodka je $\prod_{i=1}^N n_i$

Variacije

Izmed N elementov množice *razvrstimo* n elementov. Število *variacij* je $\frac{N!}{(N-n)!}$

Permutacije

Razvrstimo vse elemente množice, moči N . Množica je razdeljena na *skupine*, moči m_i , katerih elementi so nerazločljivi. Število *permutacij* je $\frac{N!}{\prod_i (m_i!)}$

Kombinacije

Izmed N elementov množice *izberemo* n elementov, oziroma izberemo podmnožico množice. Število *kombinacij brez ponavljanja* je $\binom{N}{n} = \frac{N!}{n!(N-n)!}$

Izid vsakega izmed zaporednih, neodvisnih dogodkov je element množice, moči N . Število *kombinacij s ponavljanjem* je $\binom{N+n-1}{n} = \frac{(N+n-1)!}{(N-1)!n!}$

Verjetnost

Uvedba novih slučajnih spremenljivk

Za $\mathbf{X}, \mathbf{U} \in \mathbb{R}^n$ in bijektivno $h: \mathbf{X} \rightarrow \mathbf{U}$ velja $f_{\mathbf{U}}(\mathbf{U}) = f_{\mathbf{X}}(h^{-1}(\mathbf{U})) \left| \det \left(\frac{\partial}{\partial \mathbf{U}} h^{-1}(\mathbf{U}) \right) \right|$
Če h ni bijekcija, def. bijekcije $h_i: X_i \rightarrow U \ni: \bigcup_i X_i = X$
S h_i dobimo (nenormiran) $f_{i,\mathbf{U}}$, torej $f_{\mathbf{U}}(\mathbf{U}) = \sum_i f_{i,\mathbf{U}}$

Po dogovoru je *nosilec* podan kot domena, *domena* pa je $\mathbb{R}^n$.
 $f(x_2, x_3, \dots, x_N) = \int_{x_1} f(x_1, x_2, \dots, x_N) dx_1$

Bayesov teorem in neodvisne spremenljivke

(Sledeče enačbe so simetrične)
Za dogodka A in B velja, da je verjetnost, da se zgodi dogodek A , če se zgodi dogodek B enaka $P(A \cap B) = P(B)P(A|B)$
Pri zvezni porazdelitvi s slučajnima spremenljivkama X in Y podobno velja $f_{X,Y}(x,y) = f_Y(y) f_{X|Y}(x|y)$
Dogodka A in B sta neodvisna, če
Diskretna porazdelitev: $P(A) = P(A|B) = P(A|\bar{B})$
Zvezna porazdelitev: $f_X(x) = f_{X|Y}(X|Y) = f_{X|Y}(X|\bar{Y})$
Zato za neodvisne spremenljivke velja
 $P(A,B) = P(A)P(B)$ $f_{X,Y}(X,Y) = f_X(X) f_Y(Y)$
Kvantili: vrednost x_α , za katero velja $F(x_\alpha) = \alpha$.

Zvezne porazdelitve ($x \in \mathbb{R}$)

Enakomerna porazdelitev ($X \sim U(a,b)$)

$$f_X(x) = \begin{cases} \frac{1}{b-a} & x \in (a,b) \\ 0 & \text{sicer} \end{cases} \quad F_X(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & x \in (a,b) \\ 1 & x > b \end{cases}$$

$$\bar{X} = \frac{a+b}{2} \quad \sigma_X^2 = \frac{(b-a)^2}{12} \quad \rho = 0$$

Gaussova porazdelitev ($X \sim N(\mu, \sigma)$)

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right) \quad \bar{X} = \mu \quad \sigma_X = \sigma$$
$$F_X(x) = \frac{1}{2} \left(1 + \operatorname{erf}\left(\frac{x-\mu}{\sqrt{2}\sigma}\right)\right) \quad \text{WFHM} = 2\sqrt{2\log 2}\sigma$$

2-D Gaussova porazdelitev

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}; \quad \boldsymbol{\mu} = \bar{\mathbf{X}} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}; \quad \boldsymbol{\sigma} = \begin{bmatrix} \sigma_{xx}^2 & \sigma_{yx} \\ \sigma_{xy} & \sigma_{yy}^2 \end{bmatrix}$$
$$\sigma_{ij} = \operatorname{Cov}(X_i, X_j) \Rightarrow \boldsymbol{\sigma} = \begin{bmatrix} \operatorname{Var}(X_1) & \operatorname{Cov}(X_1, X_2) \\ \operatorname{Cov}(X_1, X_2) & \operatorname{Var}(X_2) \end{bmatrix}$$
$$p_{XY}(x,y) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}}$$
$$\exp\left(-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_x)^2}{\sigma_x^2} + \frac{(y-\mu_y)^2}{\sigma_y^2} - 2\rho\frac{x-\mu_x}{\sigma_x}\frac{y-\mu_y}{\sigma_y}\right]\right)$$

Za 2-D normalno porazdelitev velja: $\rho = 0$ (spremenljivki sta nekorelirani) $\Leftrightarrow$ spremenljivki sta neodvisni

Maxwellova porazdelitev

3 neodv. in neomej. $q_i, \quad \mathbf{q} = (q_1, q_2, q_3), \quad \bar{\mathbf{q}} = 0 \quad q = |\mathbf{q}| \geq 0$

$$f_Q(q) = \sqrt{\frac{2}{\pi}} \frac{q^2}{a^3} \exp\left(-\frac{q^2}{2a^2}\right) \quad \bar{Q} = 2a\sqrt{\frac{2}{\pi}}$$
$$F_Q(q) = \operatorname{erf}\left(\frac{q}{\sqrt{2}a}\right) - \sqrt{\frac{2}{\pi}} \frac{q}{a} \exp\left(-\frac{q^2}{2a^2}\right) \quad \sigma_Q^2 = \frac{a^2(3\pi-8)}{\pi}$$

Eksponentna porazdelitev

$$t \geq 0 \quad f_T(t) = \frac{1}{\tau} \exp\left(-\frac{t}{\tau}\right) \quad F_T(t) = 1 - \exp\left(-\frac{t}{\tau}\right)$$
$$\bar{T} = \tau \quad \sigma_T^2 = \tau^2 \quad \rho = 2$$

Cauchyjeva porazdelitev ($X \sim \text{Cauchy}(\mu, \gamma)$)

$$f_X(x) = \frac{\gamma}{\pi \gamma^2 + x^2} \quad F(x) = \frac{1}{\pi} \arctan\left(\frac{(x-\mu)}{\gamma}\right) + \frac{1}{2}$$
$$\sharp(x^\alpha) \text{ za } |\alpha| \geq 1 \quad \operatorname{med}(X) = \mu \quad \operatorname{MAD}_X = \gamma$$

Studentova t-porazdelitev

$$f_T(t, \nu) = \frac{1}{\sqrt{\nu B(\frac{\nu}{2}, \frac{1}{2})}} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}} \quad \bar{T} = 0$$
$$\lim_{\nu \rightarrow \infty(30)} f_T(t, \nu) = N(0,1) \quad \sigma_T^2 = \frac{\nu}{\nu-2}$$

χ^2 porazdelitev ($X \sim \chi^2(\nu)$)

$$x > 0 \quad f_X(x, \nu) = \frac{2^{-\frac{\nu}{2}}}{\Gamma(\frac{\nu}{2})} x^{\frac{\nu}{2}-1} \exp\left(-\frac{x}{2}\right) \quad \bar{X} = \nu \quad \sigma_X^2 = 2\nu$$

F porazdelitev

$$f_F(x; \nu_1, \nu_2) = \frac{1}{B(\frac{\nu_1}{2}, \frac{\nu_2}{2})} \left(\frac{\nu_1}{\nu_2}\right)^{\frac{\nu_1}{2}} x^{\frac{\nu_1}{2}-1} \left(1 + \frac{\nu_1}{\nu_2} x\right)^{-\frac{\nu_1+\nu_2}{2}}$$
$$\bar{X} = \frac{\nu_2}{\nu_2-2} \quad \sigma_X^2 = \frac{2\nu_2^2(\nu_1+\nu_2-2)}{\nu_1(\nu_2-2)^2(\nu_2-4)}$$

Diskretne porazdelitve ($x \in \mathbb{N} \cup \{0\}$)

Bernoullijeva porazdelitev

Velja za dogodke, ki se pri poskusu zgodijo z verjetnostjo p .
 $P(X = x, p) = p^x q^{1-x} = px + q(1-x) \quad x \in \{0,1\} \quad p+q=1$

Binomska porazdelitev ($X \sim B(N, p)$)

N zaporednih, enakih, neodvisnih Bernoullijevih poskusov, P verjetnost, da je x poskusov imelo pozitiven (1) izid.

$$P(X = x, N, p) = \binom{N}{x} p^x (1-p)^{N-x}$$
$$\bar{X} = Np \quad \sigma_X^2 = Np(1-p) \quad P\left(a \leq \frac{X-\bar{X}}{\sigma_X} \leq b\right) =$$
$$= \frac{1}{\sqrt{2\pi}} \int_a^b e^{-\frac{x^2}{2}} dx = \frac{1}{2} \left(\operatorname{erf}\left(\frac{b}{\sqrt{2}}\right) - \operatorname{erf}\left(\frac{a}{\sqrt{2}}\right)\right) \quad Np, Nq > 5$$
$$M_X(t) = (1-p+pe^t)^N \quad \rho = \frac{1-2p}{(Np(1-p))^{1/2}}$$

Poissonova porazdelitev ($X \sim \text{Pois}(\lambda)$)

P verjetnost za x dogodkov, če jih pričakujemo λ .
Limita binomske $N \rightarrow \infty \ni: Np = \lambda$
 $P(X = x; \lambda) = \frac{\lambda^x}{x!} e^{-\lambda} \quad \sigma_X^2 = \bar{X} = \lambda$
 $P(X = x; \lambda) = \frac{1}{\sqrt{2\pi x}} \left(\frac{\lambda e}{x}\right)^x e^{-\lambda} \quad (\text{Stirling})$

Momenti

Surovi(algebrajski) moment:

$$M'_p = \begin{cases} \sum_X x^p f_X(x) & X \text{ diskretna} \\ \int_X x^p f_X(x) dx & X \text{ zvezna} \end{cases}$$

Centr. moment:

$$M_p = \begin{cases} \sum_X (x - \bar{X})^p f_X(x) & X \text{ diskretna} \\ \int_X (x - \bar{X})^p f_X(x) dx & X \text{ zvezna} \end{cases}$$

$$M'_0 = 1 \quad M'_1 = \bar{X} \quad M_0 = 1 \quad M_1 = 0 \quad M'_2 = M_2 + M_1'^2 = \operatorname{Var}(X) + \bar{X}^2$$

Varianca(standardna deviacija): $\operatorname{Var}(X) = \sigma_X^2 = M_2(X)$
 $\operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X)$

Poševnost: $\rho = \frac{M_3}{\sigma_X^3}; \rho < 0 \Rightarrow$ vrh desno od $\bar{X}$

Presež. splošč.: $\epsilon = \frac{M_4}{\sigma_X^4} - 3$

Kovarianca:

$$\operatorname{Cov}(X) = \sigma_{XY} = \overline{XY} - \bar{Y} \cdot \bar{X} = \overline{(X - \bar{X})(Y - \bar{Y})}$$
$$\operatorname{Var}(X \pm Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) \pm 2 \operatorname{Cov}(X, Y)$$

Lin. kor. koef.: $\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y} \in [-1,1]$

Če X, Y neodvisni, $\rho_{XY} = 0$

Mediana:

$$x_M = \operatorname{med}(X) \Leftrightarrow (P(X < x_M) \leq \frac{1}{2} \quad P(X > x_M) \leq \frac{1}{2})$$

Če je $x_{M_i} = \operatorname{med}(X)$ več, velja $\operatorname{med}(X) = \{x_{M_i}\}$

Medianski odklon: $\operatorname{MAD}(X) = \operatorname{med}(|X - \operatorname{med}(X)|)$

Momentna rodovna funkcija

M_X(t) = e^{\overline{tX}} = \sum_{k=0}^\infty t^k \frac{\overline{X^k}}{k!} \Rightarrow \overline{X^n} = \frac{d^n}{dt^n} M_X(t) \Big|_{t=0}

Rodovna funkcija verjetnosti

G_x(z) = z^{\overline{X}}; |z| < 1 \Rightarrow \overline{x^n} = \left(z \frac{d}{dz}\right)^n G_X(z) \Big|_{z=1}

Karakteristična funkcija

\Phi_X(t) = e^{it\overline{X}}; t \in \mathbb{R} \Rightarrow \overline{X^n} = \frac{1}{i^n} \frac{d^n}{dt^n} \Phi_X(t) \Big|_{t=0}

Konvolucija

X, Y neodv., zvezni, Z = X + Y \implies f_Z(z) = f_X(x) * f_Y(y)
X, Y neodv., diskretni, P(Z = n) = \sum_i P(X = i)P(Y = n - i)

N(\mu_1, \sigma_1) + N(\mu_2, \sigma_2) \sim N(\mu_1 + \mu_2, \sigma_1 + \sigma_2)
Cauchy(\mu_1, \gamma_1) + Cauchy(\mu_2, \gamma_2) \sim Cauchy(\mu_1 + \mu_2, \gamma_1 + \gamma_2)
\sum_{i=1}^\nu (N(0, 1))^2 \sim \chi^2(\nu) \qquad \chi^2(\nu_1) + \chi^2(\nu_2) \sim \chi^2(\nu_1 + \nu_2)

B(N_1, p) + B(N_2, p) \sim B(N_1 + N_2, p)
Pois(\lambda_1) + Pois(\lambda_2) \sim Pois(\lambda_1 + \lambda_2)

Centralni limitni izrek

Izrojena porazdelitev je podprta skoraj nikjer. Za neizrojeno f_X (vse omenjene porazdelitve), stabilno porazdelitev g_X (Gaussova ali Cauchyjeva) velja f_X * f_X * \cdots * f_X = g_X

X_1, ..., X_N neodvisne z enako porazdelitvijo;
\mu_x = \mu_1 = \dots = \mu_N < \infty; \sigma_x = \sigma_1 = \dots = \sigma_N < \infty.
Definiramo Y_n = X_1 + \dots + X_n; \quad Z_n = \frac{Y_n - \overline{Y_n}}{\sigma_{Y_n}}. Tedaj Z_n v limiti n \to \infty postane normalna porazdelitev
Z, \quad p_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.

Statistika

S populacija (\mu, \sigma) moči N, S_n vzorec moči n, \forall X_i \in S_n \subset S
Stat. in cenilka: \theta = T(S) \qquad \theta_n = T(S_n)
n \to \infty \qquad T(S_n) = T(S) \qquad \text{Var}(S_n) = 0 \qquad (\text{doslednost})
\forall n \qquad E(S_n) = S \qquad (\text{nepistranskost})

Vzorčno povprečje (cenilka): \overline{X} = \frac{1}{n} \sum_{i=1}^n X_i
E(\overline{X}) = \mu \qquad \text{Var}(\overline{X}) = \frac{N-n}{N-1} \frac{\sigma^2}{n} \to \frac{\sigma^2}{n} \text{ (z nadom. / } N \gg 1)
Vzorčna varianca (cenilka): S_X^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \overline{X})^2
E(S_X^2) = \sigma^2 \qquad \text{Var}(S_X^2) = \frac{2\sigma^4}{n}

Za cenilki X_1 iz populacije z \mu_1, \sigma_1 in X_2 iz populacije z \mu_2, \sigma_2 ter Z = X_1 \pm X_2 velja E(\overline{Z}) = \mu_1 \pm \mu_2
Var(\overline{Z}) = Var(\overline{X_1}) + Var(\overline{X_2}) \qquad \overline{Z} \sim N\left(E(\overline{Z}), \sqrt{\text{Var}(\overline{Z})}\right)
j = k \frac{\sigma}{\sqrt{N}} (k \in \mathbb{N}), \quad CL = P(M - j < \mu < M + j)

T-test: T = \frac{M - \mu_0}{s/\sqrt{N}}, Z-test: Z = \frac{M - \mu_0}{\sigma/\sqrt{N}};
s = \sqrt{\frac{1}{N-1} \sum_{i=1}^n (x_i - m)^2}. Če N \geq 30, uporabimo Gaussovo porazdelitev, sicer Studentovo.

Matematičen dodatek

Binomski koeficient ter formula
{N \choose n} = \frac{N!}{n!(N-n)!} \qquad (x + y)^N = \sum_{k=0}^N {N \choose k} x^k y^{N-k}

Funkcija napake: \text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x \exp(-t^2) dt

Jacobijeva matrika totalnega odvoda:
f, \mathbf{x} \in \mathbb{R}^n \implies \frac{\partial \mathbf{f}}{\partial \mathbf{x}} = \left[\frac{\partial f_i}{\partial x_j} \right]_{ij}

Stirlingova formula: n \to \infty (> 8) \qquad n! = \sqrt{2\pi n} \left(\frac{n}{e}\right)^n

Konvolucijski integral: (f * g)(t) = \int_{-\infty}^\infty f(\tau)g(t - \tau) d\tau